

Initiating Search

November 10, 2025, 9:22 PM

• Search:

CAS Lexicon Search:

Groebke-Blackburn-Bienayme reaction - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (191)	 References	View Results
Exported: Retrieved Related Reaction Results + Filters (472)	 Reactions	View Results
Filtered By:		
Yield: 90-100%, 80-89%, 70-79%		

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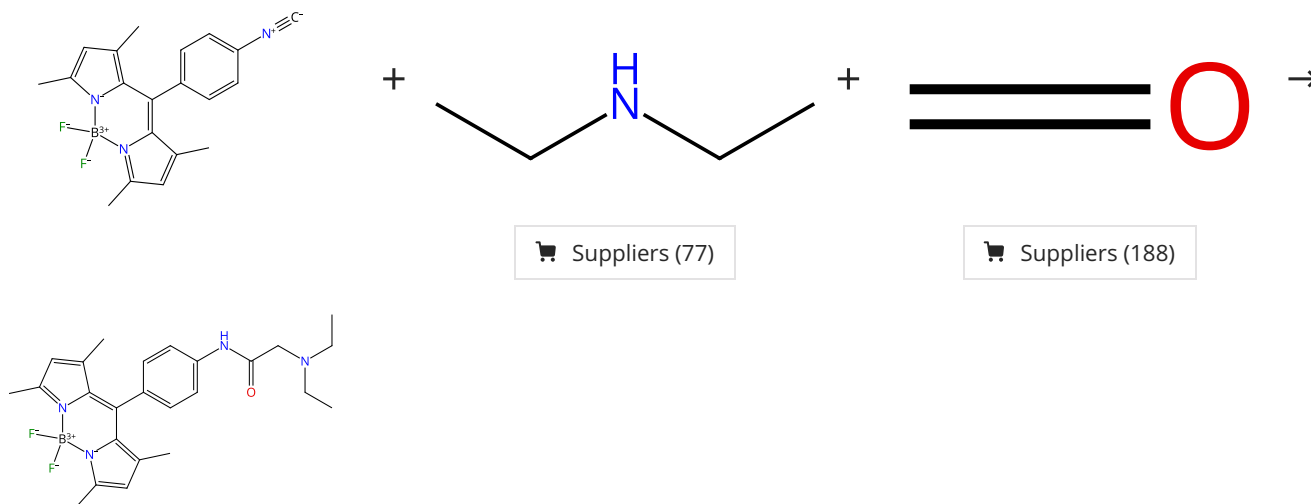
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Reactions (50)

[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 99%



31-614-CAS-28375570

Steps: 1 Yield: 99%

Multicomponent Reactions for de Novo Synthesis of BODIPY Probes: In Vivo Imaging of Phagocytic Macrophages

By: Vazquez-Romero, Ana; et al

Journal of the American Chemical Society (2013), 135(43), 16018-16021.

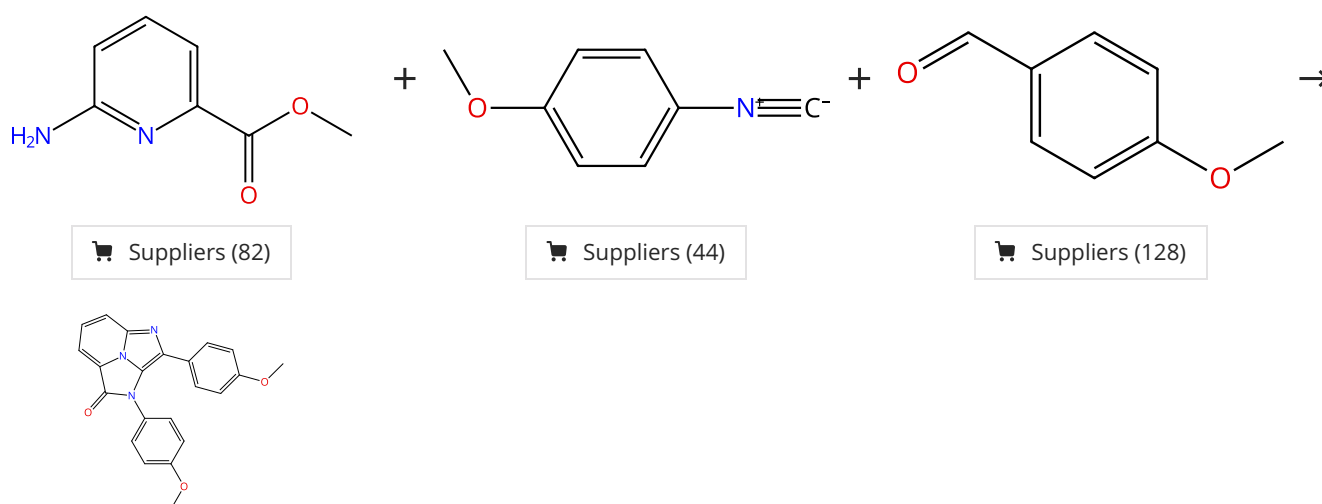
1.1 Reagents: Formic acid

Solvents: Chloroform, *tert*-Butanol, Water; 16 h, rt

Experimental Protocols

Scheme 2 (1 Reaction)

Steps: 1 Yield: 99%



31-614-CAS-29427742

Steps: 1 Yield: 99%

Facile Synthesis of New Functionalized Triazacyclopenta[cd]indenones Through a Multicomponent Reaction Followed by an Intramolecular Aminolysis

By: Shen, Guo-Bing; et al

Journal of Heterocyclic Chemistry (2018), 55(4), 814-820.

1.1 Catalysts: Ytterbium triflate

Solvents: 1-Butanol; 6 h, 55 °C; 55 °C → rt

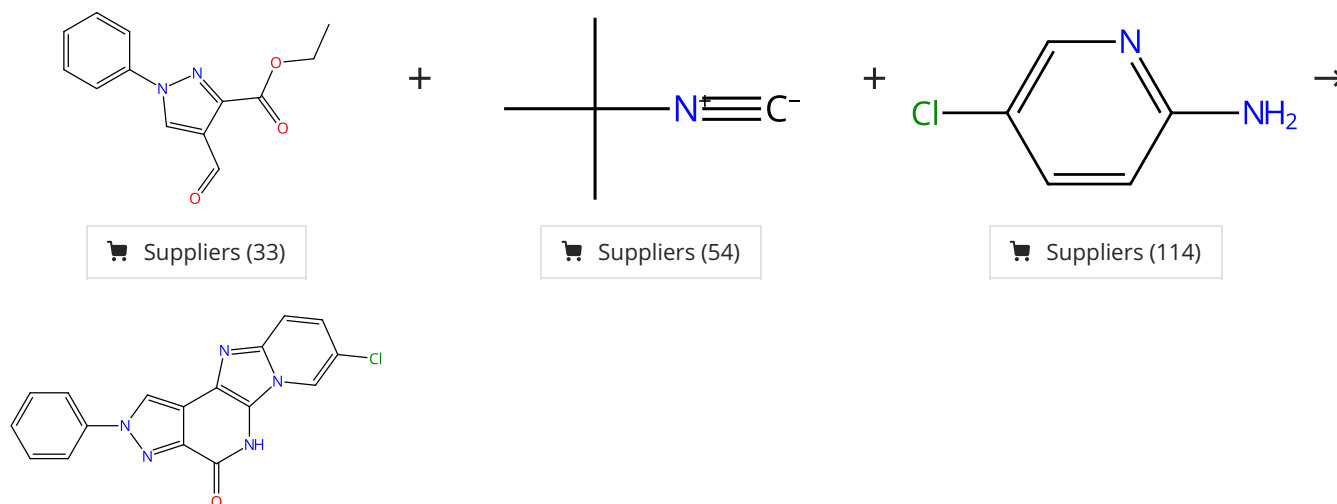
1.2 Reagents: Potassium carbonate

Solvents: 1-Butanol; 6 h, 100 °C

Experimental Protocols

Scheme 3 (1 Reaction)

Steps: 1 Yield: 99%



31-614-CAS-29795393

Steps: 1 Yield: 99%

In(OTf)₃-HBF₄ Assisted Multicomponent Approach for One-Pot Synthesis of Pyrazolopyridinone Fused Imidazopyridines

By: Devi, Nisha; et al

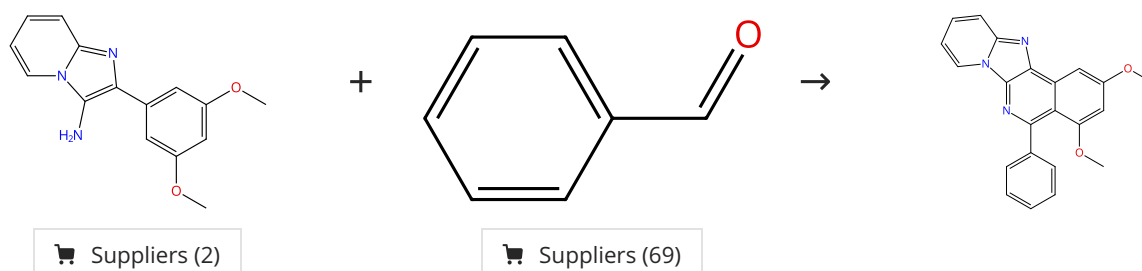
ChemistrySelect (2016), 1(15), 4696-4703.

- 1.1 **Catalysts:** Indium triflate
Solvents: Toluene; 5 min, 120 °C
- 1.2 **Reagents:** Tetrafluoroboric acid
Solvents: Water; 135 min, 120 °C
- 1.3 **Solvents:** Ethyl acetate; 10 min, rt

Experimental Protocols

Scheme 4 (1 Reaction)

Steps: 1 Yield: 99%



31-614-CAS-37923774

Steps: 1 Yield: 99%

The full spectrum tuning of fluorescent molecules via a one-pot multicomponent reaction

By: Bedard, Nathan; et al

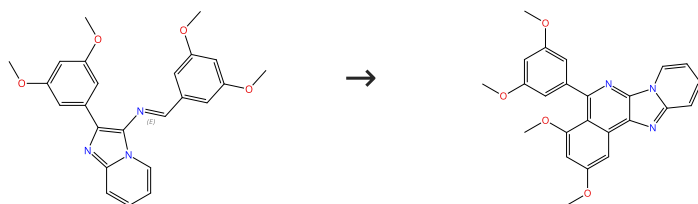
Tetrahedron Letters (2023), 130, 154748.

- 1.1 **Catalysts:** Copper(II) triflate
Solvents: Dichloromethane; 20 min, 100 °C

Experimental Protocols

Scheme 5 (1 Reaction)

Steps: 1 Yield: 99%

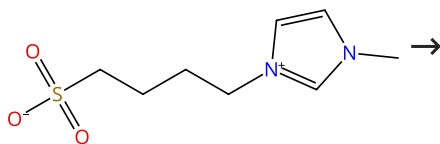


Double bond geometry shown

31-614-CAS-37923777	Steps: 1 Yield: 99%	The full spectrum tuning of fluorescent molecules via a one-pot multicomponent reaction
1.1 Catalysts: Copper(II) triflate Solvents: 1,2-Dichloroethane; 20 min, rt → 120 °C		By: Bedard, Nathan; et al
Experimental Protocols		Tetrahedron Letters (2023), 130, 154748.

Scheme 6 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (41)

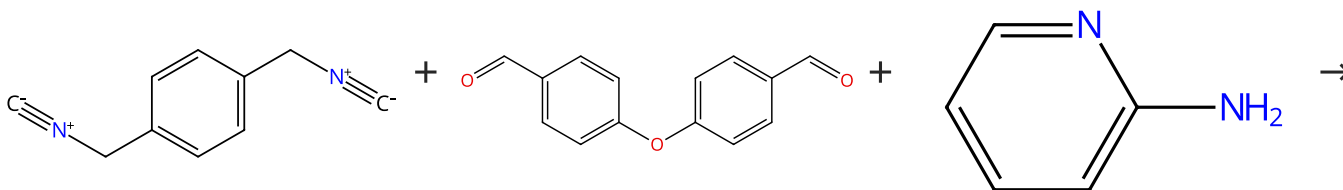
Multi-component
structure image
available in CAS
SciFinder

Suppliers (50)

31-614-CAS-34387877	Steps: 1 Yield: 99%	Groebke-Blackburn-Bienayme Multicomponent Reaction Catalysed by Reusable Bronsted acidic ionic Liquids
1.1 Reagents: Sulfuric acid Solvents: Water; 12 h, 60 °C		By: Anjos, Nicolas S.; et al
Experimental Protocols		European Journal of Organic Chemistry (2022), 2022(40), e202200615.

Scheme 7 (1 Reaction)

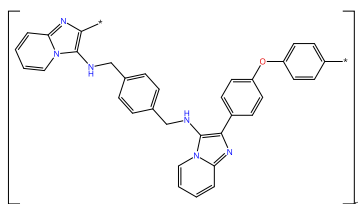
Steps: 1 Yield: 98%



Suppliers (9)

Suppliers (62)

Suppliers (89)



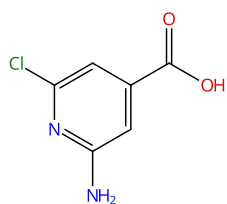
31-614-CAS-34585965	Steps: 1 Yield: 98%	Multicomponent Polymerizations of Isocyanides, Aldehydes, and 2-Aminopyridine toward Imidazo[1,2-a]pyridine- Containing Fused Heterocyclic Polymers
1.1 Catalysts: <i>p</i> -Toluenesulfonic acid Solvents: Ethanol; 24 h, 70 °C		By: Chen, Tingting; et al
Experimental Protocols		Macromolecules (Washington, DC, United States) (2022), 55(19), 8590-8598.

Scheme 8 (1 Reaction)

Steps: 1 Yield: 98%

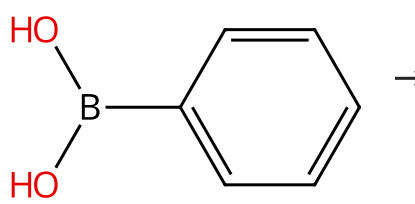
3053724-30-1

Image Not Available +



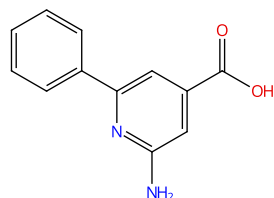
Suppliers (61)

+



Suppliers (127)

→



DNA-encoded library conjugates

31-614-CAS-41796215

Steps: 1 Yield: 98%

1.1 **Reagents:** Diisopropylethylamine, *O*-(7-Azabenzotriazol-1-yl)-*N,N,N,N*-tetramethyluronium hexafluorophosphate
Solvents: Dimethylacetamide, Water; overnight, pH 9.5, 40 °C

1.2 **Reagents:** Cesium hydroxide
Catalysts: Palladate(1-), [2'-(amino-κ*N*)[1,1'-biphenyl]-2-yl-κ*C*]chloro[2'-(dicyclohexylphosphino-κ*P*)-2,6-dimethoxy[1,1'-biphenyl]-3-sulfonato]-, sodium (1:1)
Solvents: 1,4-Dioxane, Dimethylacetamide, Water; 15 min, 80 °C; 10 min, rt

1.3 **Reagents:** Sodium diethyldithiocarbamate
Solvents: Water; 30 min, 60 °C

Experimental Protocols

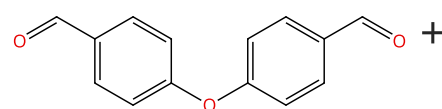
Groebke-Blackburn-Bienayme Reaction for DNA-Encoded Library Technology

By: An, Yujin; et al

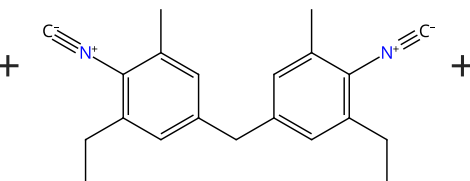
Organic Letters (2023), 25(24), 4445-4450.

Scheme 9 (1 Reaction)

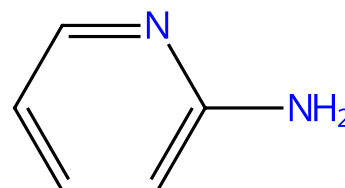
Steps: 1 Yield: 98%



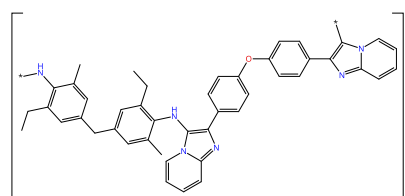
Suppliers (62)



Supplier (1)



Suppliers (89)



31-614-CAS-34585957

Steps: 1 Yield: 98%

1.1 **Catalysts:** *p*-Toluenesulfonic acid
Solvents: Ethanol; 24 h, 70 °C

Experimental Protocols

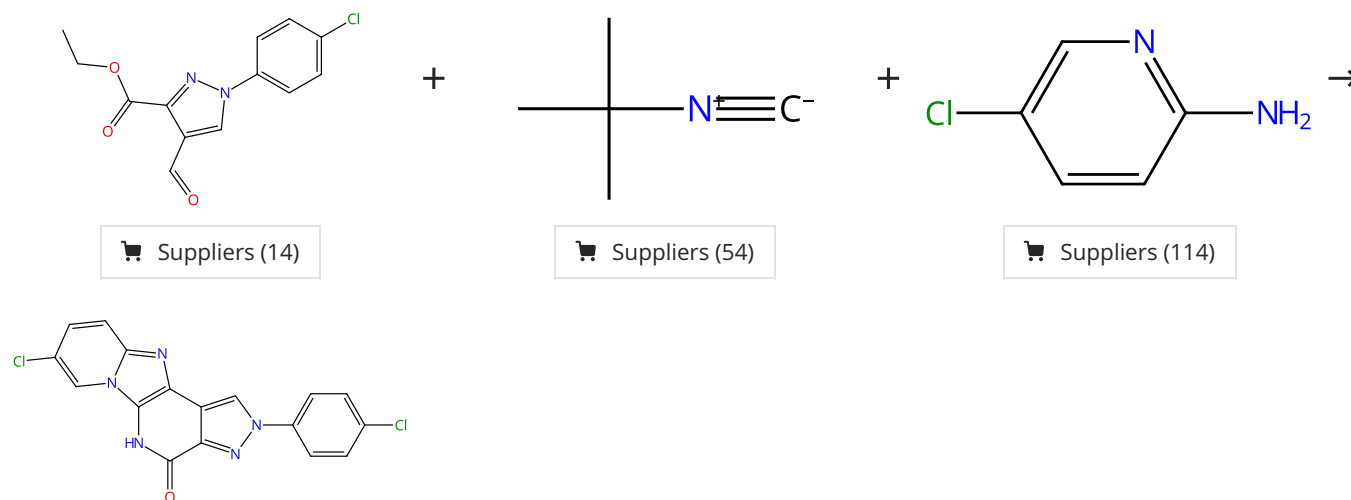
Multicomponent Polymerizations of Isocyanides, Aldehydes, and 2-Aminopyridine toward Imidazo[1,2-*a*]pyridine-Containing Fused Heterocyclic Polymers

By: Chen, Tingting; et al

Macromolecules (Washington, DC, United States) (2022), 55(19), 8590-8598.

Scheme 10 (1 Reaction)

Steps: 1 Yield: 98%



31-614-CAS-25711299

Steps: 1 Yield: 98%

In(OTf)₃-HBF₄ Assisted Multicomponent Approach for One-Pot Synthesis of Pyrazolopyridinone Fused Imidazopyridines

By: Devi, Nisha; et al

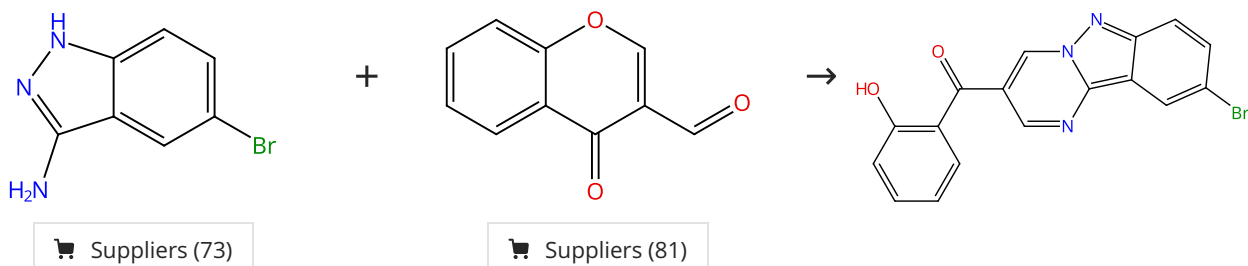
ChemistrySelect (2016), 1(15), 4696-4703.

- 1.1 **Catalysts:** Indium triflate
Solvents: Toluene; 5 min, 120 °C
- 1.2 **Reagents:** Tetrafluoroboric acid
Solvents: Water; 135 min, 120 °C

Experimental Protocols

Scheme 11 (1 Reaction)

Steps: 1 Yield: 98%



31-614-CAS-24938803

Steps: 1 Yield: 98%

An approach towards the synthesis of novel fused nitrogen tricyclic heterocyclic scaffolds via GBB reaction

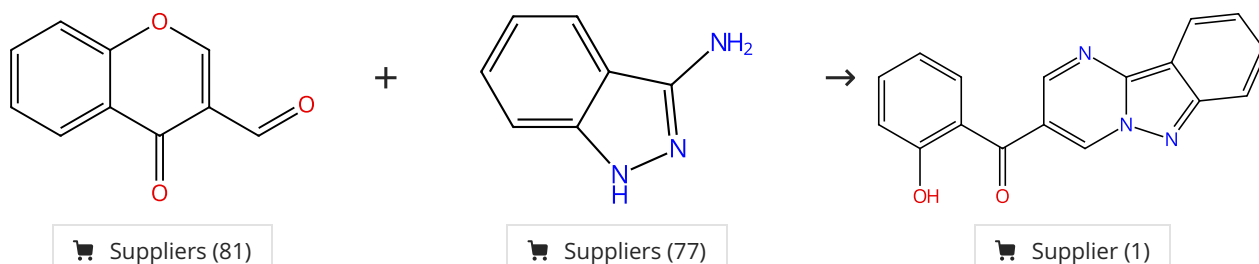
By: Balwe, Sandip Gangadhar; et al

Organic & Biomolecular Chemistry (2018), 16(8), 1287-1296.

- 1.1 **Catalysts:** Acetic acid
Solvents: Ethanol; 1 h, rt
- Experimental Protocols

Scheme 12 (1 Reaction)

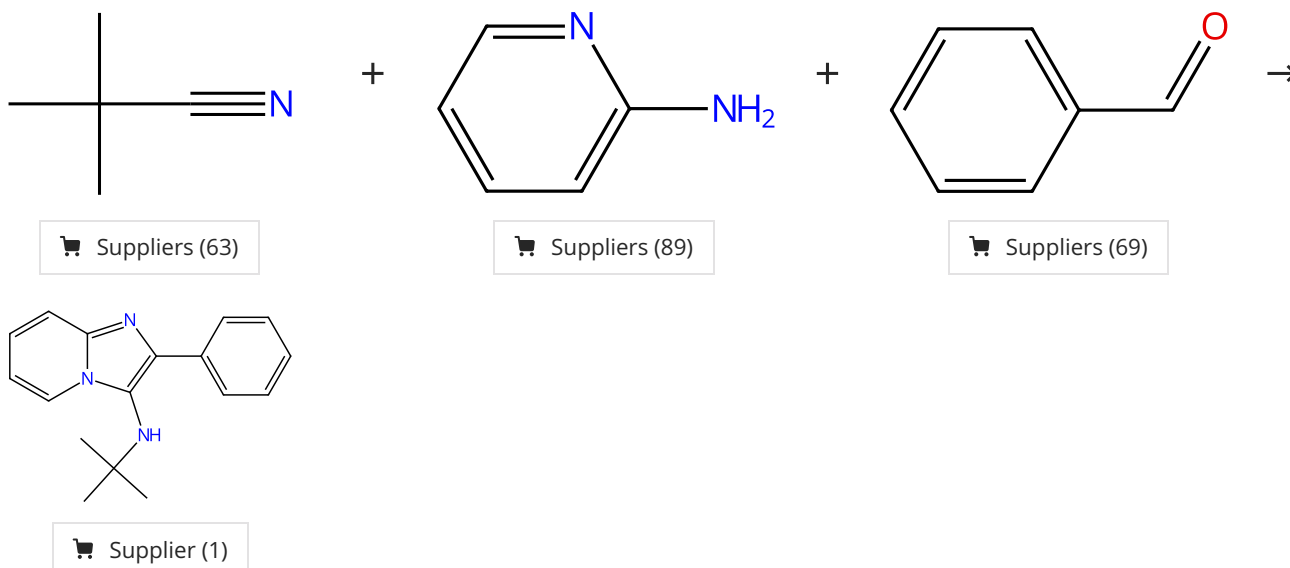
Steps: 1 Yield: 97%



31-614-CAS-29817755	Steps: 1 Yield: 97%	An approach towards the synthesis of novel fused nitrogen tricyclic heterocyclic scaffolds via GBB reaction
1.1 Catalysts: Acetic acid Solvents: Ethanol; 0.5 h, rt		By: Balwe, Sandip Gangadhar; et al
Experimental Protocols		Organic & Biomolecular Chemistry (2018), 16(8), 1287-1296.

Scheme 13 (1 Reaction)

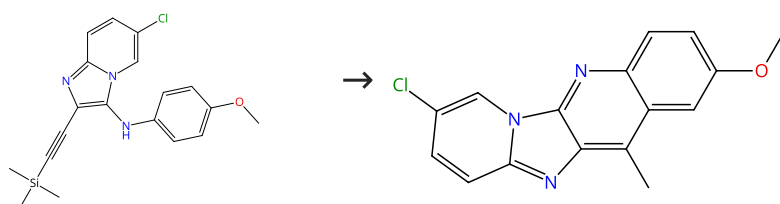
Steps: 1 Yield: 97%



31-614-CAS-25031025	Steps: 1 Yield: 97%	Visible-light-activated C-C and C-N bond formation in the synthesis of imidazo[1,2-a]pyridines and imidazo[2,1-b]thiazoles under catalyst and solvent-free conditions
1.1 3 h		By: Shivhare, Km Neha; et al
Experimental Protocols		New Journal of Chemistry (2018), 42(20), 16591-16601.

Scheme 14 (1 Reaction)

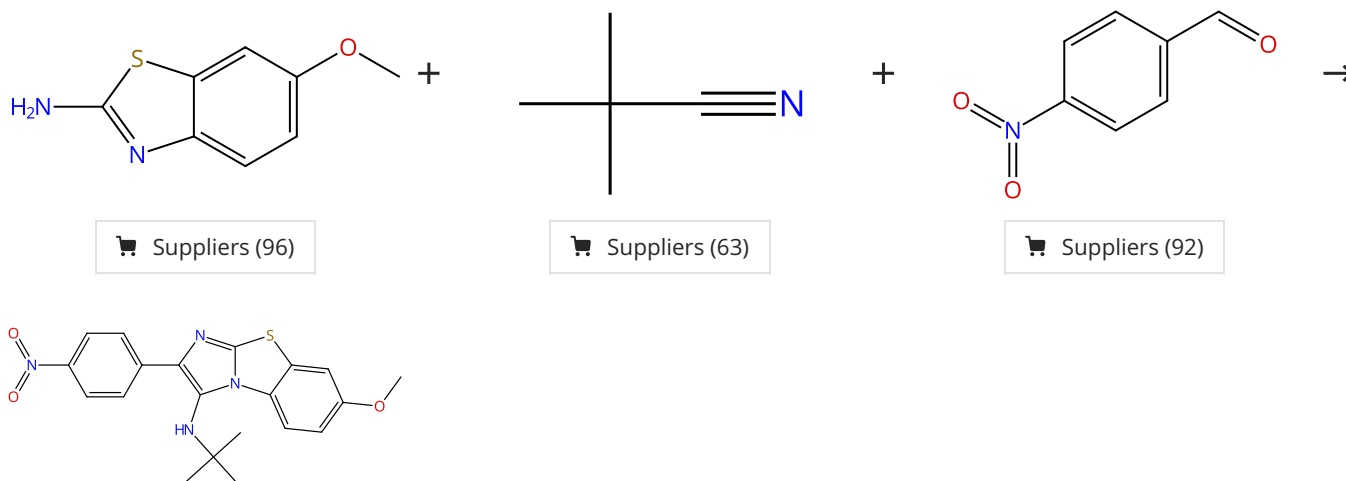
Steps: 1 Yield: 97%



31-614-CAS-28393242	Steps: 1 Yield: 97%	Efficient Metal-Free Synthesis of Various Pyrindo [2',1':2,3]imidazo- [4,5-b]quinolines
1.1 Reagents: 1,8-Diazabicyclo[5.4.0]undec-7-ene Solvents: 1,4-Dioxane; 1 h, 220 °C		By: Arnould, Mathieu; et al
Experimental Protocols		Chemistry - A European Journal (2013), 19(37), 12249-12253.

Scheme 15 (1 Reaction)

Steps: 1 Yield: 97%



31-614-CAS-29081516

Steps: 1 Yield: 97%

Visible-light-activated C-C and C-N bond formation in the synthesis of imidazo[1,2-a]pyridines and imidazo[2,1-b]thiazoles under catalyst and solvent-free conditions

1.1 2 h

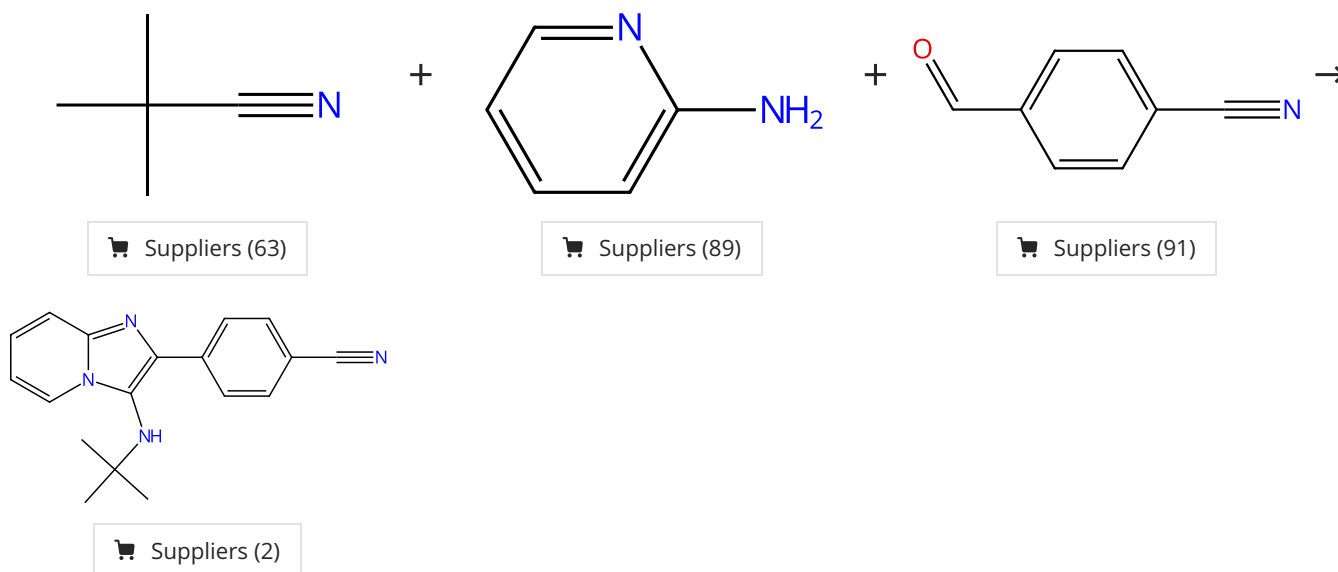
Experimental Protocols

By: Shivhare, Km Neha; et al

New Journal of Chemistry (2018), 42(20), 16591-16601.

Scheme 16 (1 Reaction)

Steps: 1 Yield: 97%



31-614-CAS-25900104

Steps: 1 Yield: 97%

Visible-light-activated C-C and C-N bond formation in the synthesis of imidazo[1,2-a]pyridines and imidazo[2,1-b]thiazoles under catalyst and solvent-free conditions

1.1 2.5 h

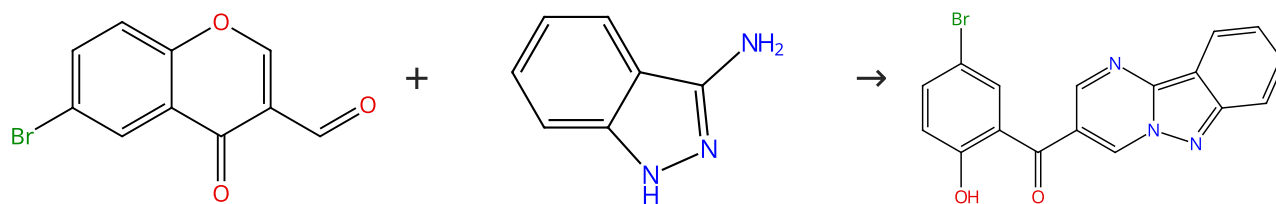
Experimental Protocols

By: Shivhare, Km Neha; et al

New Journal of Chemistry (2018), 42(20), 16591-16601.

Scheme 17 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (78)

Suppliers (77)

31-614-CAS-27969969

Steps: 1 Yield: 97%

An approach towards the synthesis of novel fused nitrogen tricyclic heterocyclic scaffolds via GBB reaction

By: Balwe, Sandip Gangadhar; et al

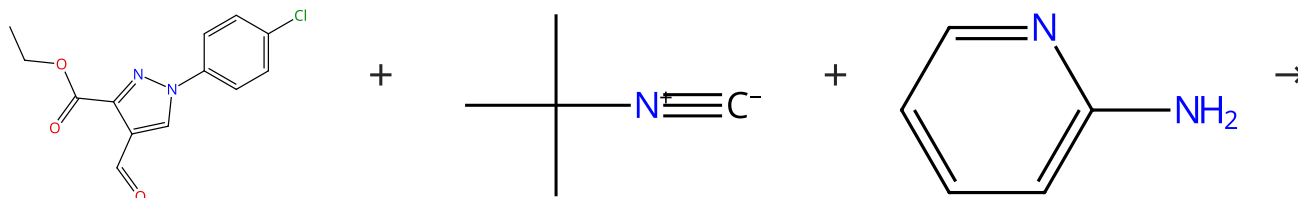
Organic & Biomolecular Chemistry (2018), 16(8), 1287-1296.

1.1 Catalysts: Acetic acid
Solvents: Ethanol; 0.5 h, rt

Experimental Protocols

Scheme 18 (1 Reaction)

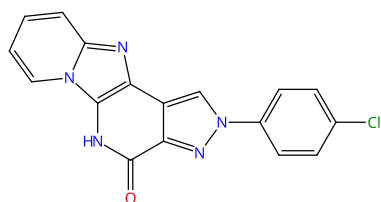
Steps: 1 Yield: 97%



Suppliers (14)

Suppliers (54)

Suppliers (89)



31-614-CAS-29859556

Steps: 1 Yield: 97%

In(OTf)₃-HBF₄ Assisted Multicomponent Approach for One-Pot Synthesis of Pyrazolopyridinone Fused Imidazopyridines

By: Devi, Nisha; et al

ChemistrySelect (2016), 1(15), 4696-4703.

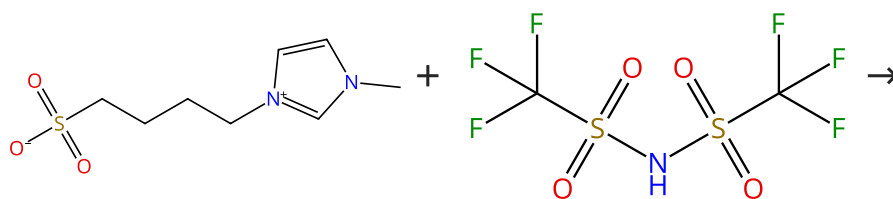
1.1 Catalysts: Indium triflate
Solvents: Toluene; 5 min, 120 °C

1.2 Reagents: Tetrafluoroboric acid
Solvents: Water; 135 min, 120 °C

Experimental Protocols

Scheme 19 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (41)

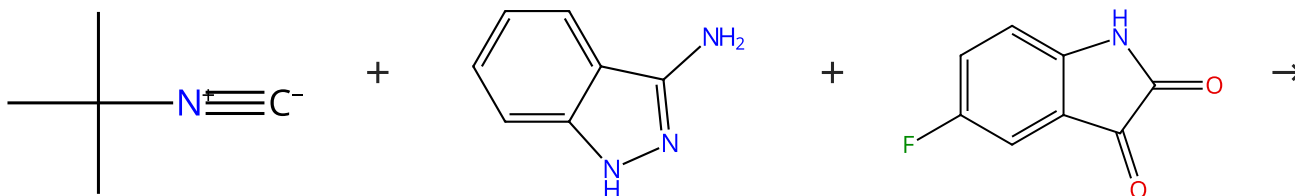
Suppliers (69)

Multi-component structure image available in CAS SciFinder

Suppliers (44)

31-614-CAS-34387885 Steps: 1 Yield: 97% 1.1 Solvents: Methanol; 0 °C; 24 h, rt	Groebe-Blackburn-Bienayme Multicomponent Reaction Catalysed by Reusable Bronsted acidic ionic Liquids By: Anjos, Nicolas S.; et al European Journal of Organic Chemistry (2022), 2022(40), e202200615.
Experimental Protocols	

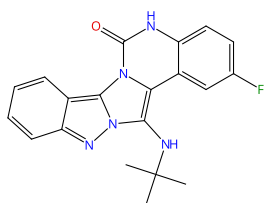
Scheme 20 (1 Reaction)

Steps: **1** Yield: **97%**

Suppliers (54)

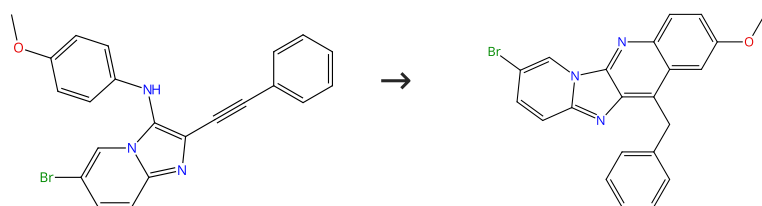
Suppliers (77)

Suppliers (112)



31-614-CAS-27517961 Steps: 1 Yield: 97% 1.1 Catalysts: β -Cyclodextrin, 6 ^A -(hydrogen butanedioate) Solvents: Water; 30 min, 100 °C	Succinyl-β-cyclodextrin-driven synthesis of a nitrogen-fused five-ring heterocycle using GBB-based [4 + 1] cycloaddition via supramolecular host-guest interactions By: Shinde, Vijay Vilas; et al Tetrahedron (2019), 75(6), 778-783.
Experimental Protocols	

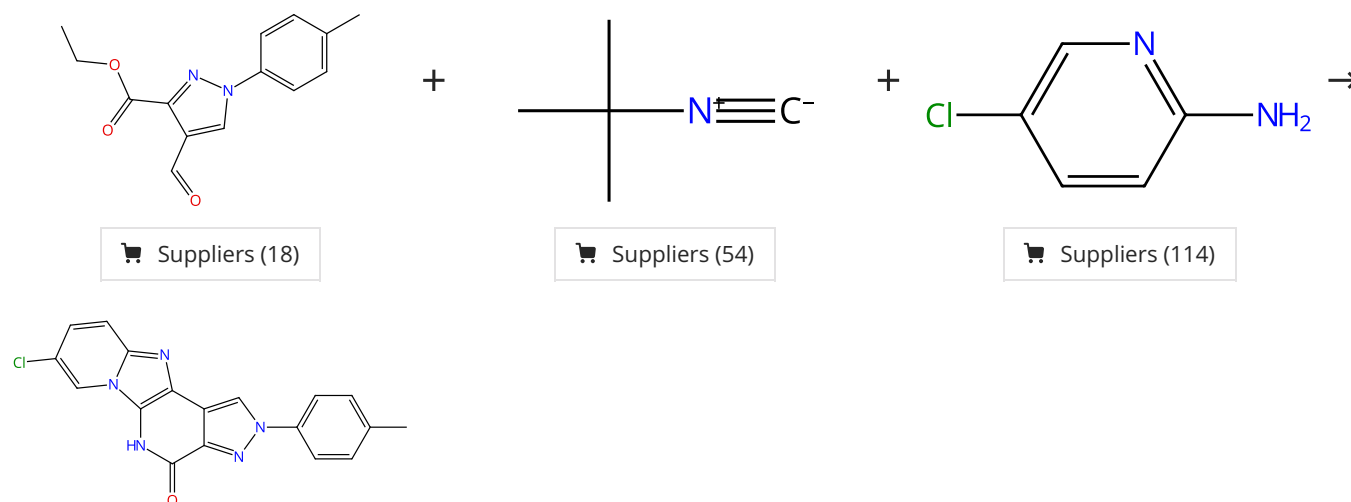
Scheme 21 (1 Reaction)

Steps: **1** Yield: **97%**

31-614-CAS-25789306 Steps: 1 Yield: 97% 1.1 Reagents: 1,8-Diazabicyclo[5.4.0]undec-7-ene Solvents: 1,4-Dioxane; 5 min, 220 °C	Efficient Metal-Free Synthesis of Various Pyrido [2',1':2,3] imidazo- [4,5-b]quinolines By: Arnould, Mathieu; et al Chemistry - A European Journal (2013), 19(37), 12249-12253.
Experimental Protocols	

Scheme 22 (1 Reaction)

Steps: 1 Yield: 96%



31-614-CAS-30416272

Steps: 1 Yield: 96%

In(OTf)₃-HBF₄ Assisted Multicomponent Approach for One-Pot Synthesis of Pyrazolopyridinone Fused Imidazopyridines

By: Devi, Nisha; et al

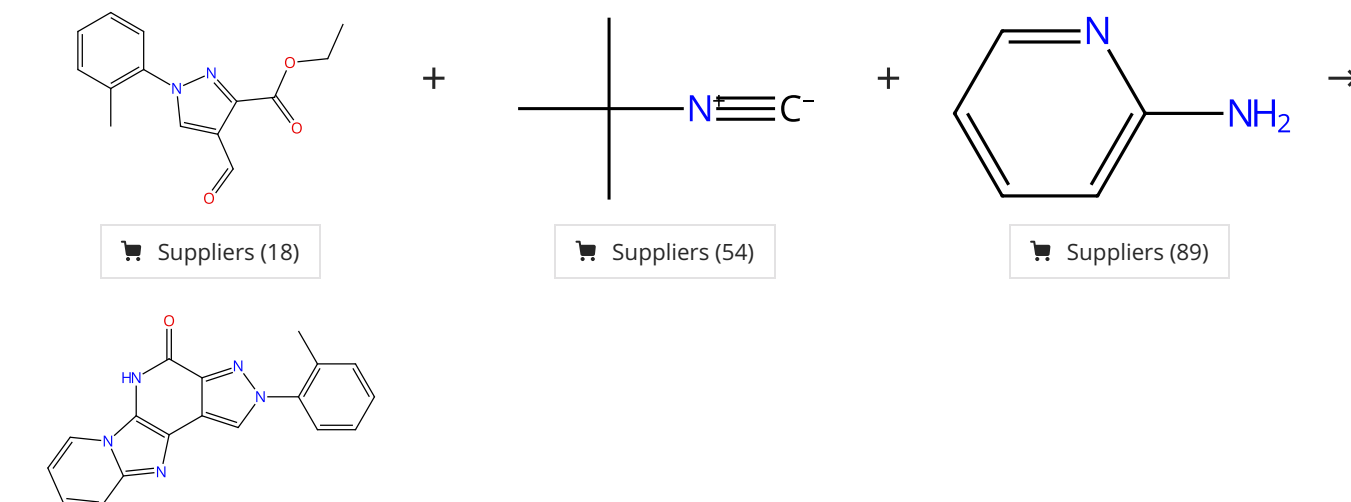
ChemistrySelect (2016), 1(15), 4696-4703.

- 1.1 **Catalysts:** Indium triflate
Solvents: Toluene; 5 min, 120 °C
- 1.2 **Reagents:** Tetrafluoroboric acid
Solvents: Water; 135 min, 120 °C

Experimental Protocols

Scheme 23 (1 Reaction)

Steps: 1 Yield: 96%



31-614-CAS-24976198

Steps: 1 Yield: 96%

In(OTf)₃-HBF₄ Assisted Multicomponent Approach for One-Pot Synthesis of Pyrazolopyridinone Fused Imidazopyridines

By: Devi, Nisha; et al

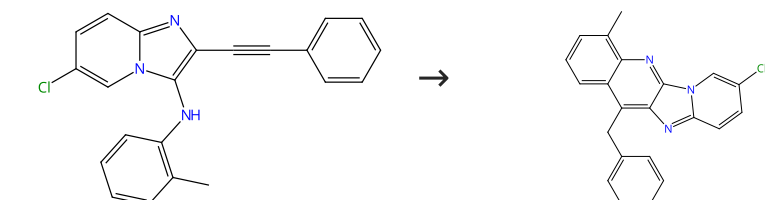
ChemistrySelect (2016), 1(15), 4696-4703.

- 1.1 **Catalysts:** Indium triflate
Solvents: Toluene; 5 min, 120 °C
- 1.2 **Reagents:** Tetrafluoroboric acid
Solvents: Water; 135 min, 120 °C

Experimental Protocols

Scheme 24 (1 Reaction)

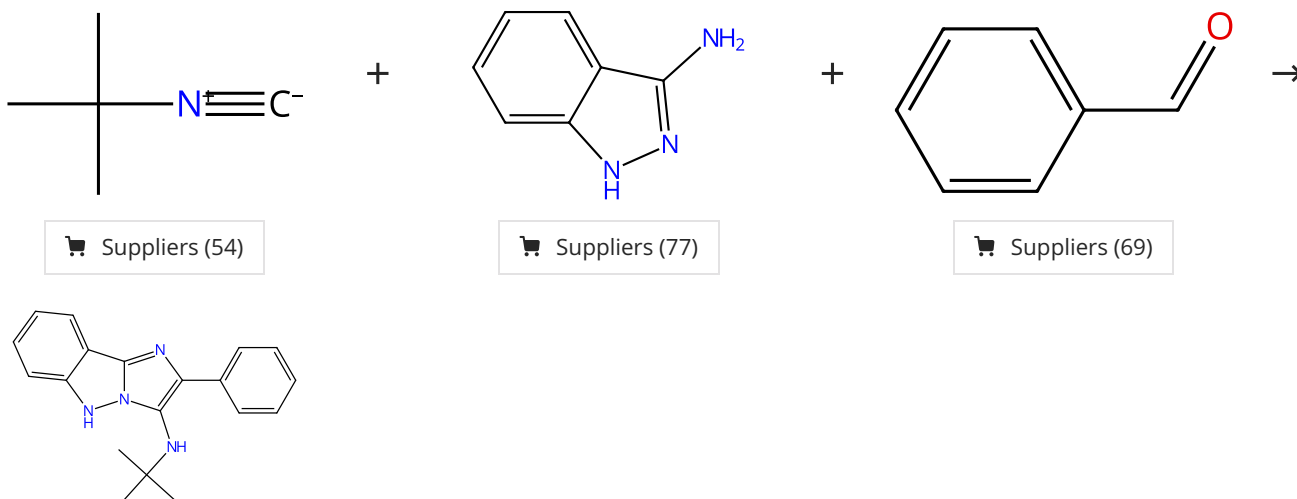
Steps: 1 Yield: 96%



31-614-CAS-29195378	Steps: 1 Yield: 96%	Efficient Metal-Free Synthesis of Various Pyrido [2',1':2,3] imidazo- [4,5-b]quinolines
1.1 Reagents: 1,8-Diazabicyclo[5.4.0]undec-7-ene Solvents: 1,4-Dioxane; 5 min, 220 °C		By: Arnould, Mathieu; et al
Experimental Protocols		Chemistry - A European Journal (2013), 19(37), 12249-12253.

Scheme 25 (1 Reaction)

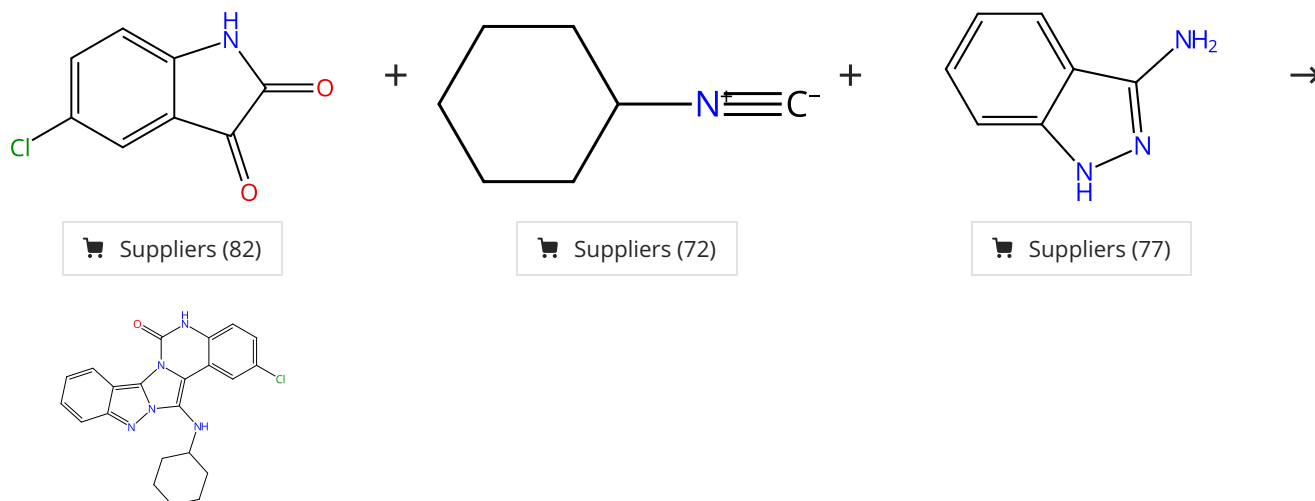
Steps: 1 Yield: 96%



31-614-CAS-25333674	Steps: 1 Yield: 96%	An approach towards the synthesis of novel fused nitrogen tricyclic heterocyclic scaffolds via GBB reaction
1.1 Catalysts: Acetic acid Solvents: Ethanol; 0.5 h, rt		By: Balwe, Sandip Gangadhar; et al
Experimental Protocols		Organic & Biomolecular Chemistry (2018), 16(8), 1287-1296.

Scheme 26 (1 Reaction)

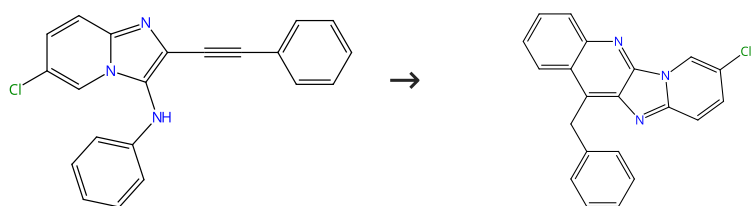
Steps: 1 Yield: 96%



31-614-CAS-26661719	Steps: 1 Yield: 96%	Succinyl-β-cyclodextrin-driven synthesis of a nitrogen-fused five-ring heterocycle using GBB-based [4 + 1] cycloaddition via supramolecular host-guest interactions
1.1 Catalysts: β-Cyclodextrin, 6 ^A -(hydrogen butanedioate) Solvents: Water; 30 min, 100 °C		By: Shinde, Vijay Vilas; et al
Experimental Protocols		Tetrahedron (2019), 75(6), 778-783.

Scheme 27 (1 Reaction)

Steps: 1 Yield: 96%



31-614-CAS-27706798

Steps: 1 Yield: 96%

Efficient Metal-Free Synthesis of Various Pyrido [2',1':2,3] imidazo- [4,5-b]quinolines

By: Arnould, Mathieu; et al

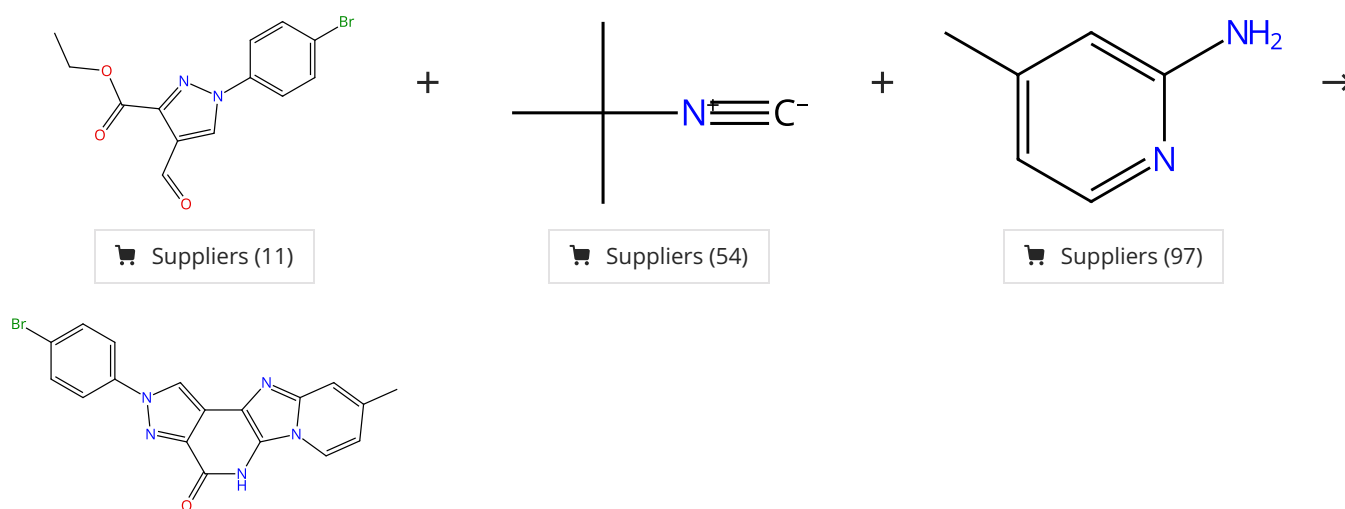
Chemistry - A European Journal (2013), 19(37), 12249-12253.

1.1 Reagents: 1,8-Diazabicyclo[5.4.0]undec-7-ene
Solvents: 1,4-Dioxane; 5 min, 220 °C

Experimental Protocols

Scheme 28 (1 Reaction)

Steps: 1 Yield: 96%



31-614-CAS-29181714

Steps: 1 Yield: 96%

In(OTf)₃-HBF₄ Assisted Multicomponent Approach for One-Pot Synthesis of Pyrazolopyridinone Fused Imidazopyridines

By: Devi, Nisha; et al

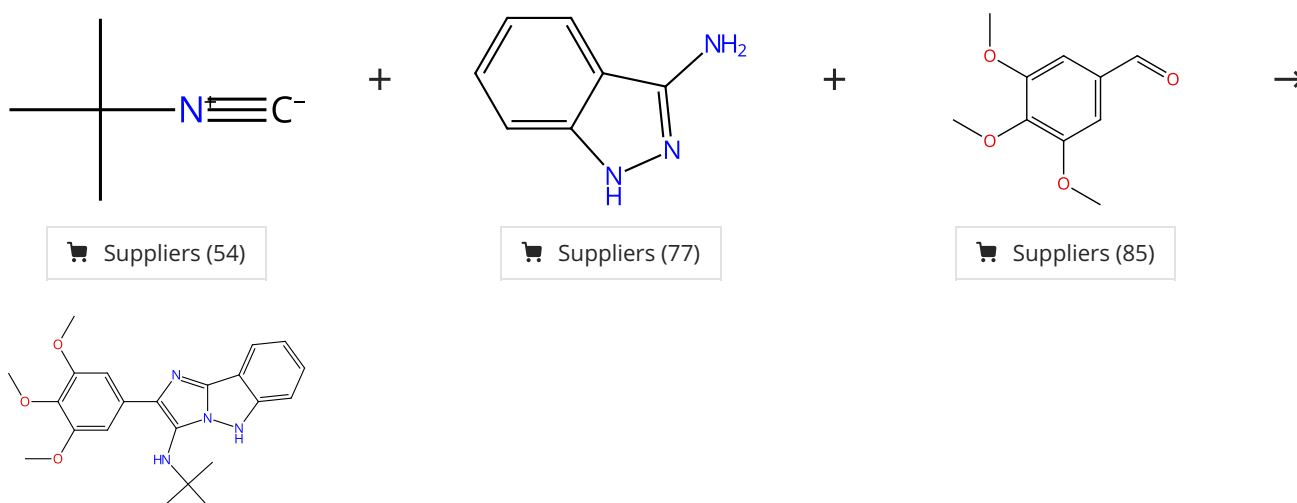
ChemistrySelect (2016), 1(15), 4696-4703.

1.1 Catalysts: Indium triflate
Solvents: Toluene; 5 min, 120 °C
1.2 Reagents: Tetrafluoroboric acid
Solvents: Water; 135 min, 120 °C

Experimental Protocols

Scheme 29 (1 Reaction)

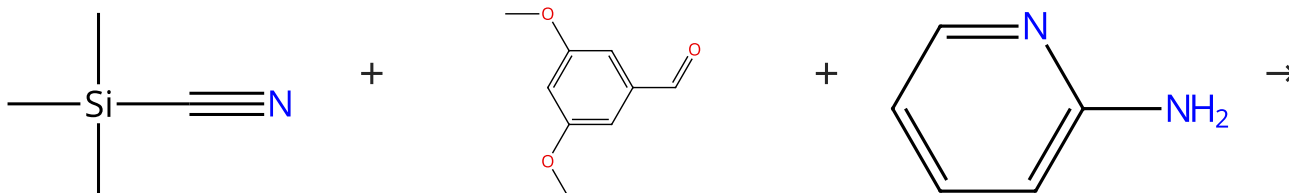
Steps: 1 Yield: 96%



31-614-CAS-29958353	Steps: 1 Yield: 96%	An approach towards the synthesis of novel fused nitrogen tricyclic heterocyclic scaffolds via GBB reaction
1.1 Catalysts: Acetic acid Solvents: Ethanol; 1.5 h, rt		By: Balwe, Sandip Gangadhar; et al
Experimental Protocols		Organic & Biomolecular Chemistry (2018), 16(8), 1287-1296.

Scheme 30 (1 Reaction)

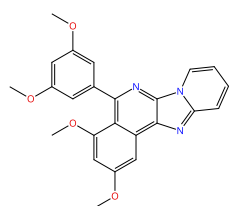
Steps: 1 Yield: 96%



Suppliers (43)

Suppliers (93)

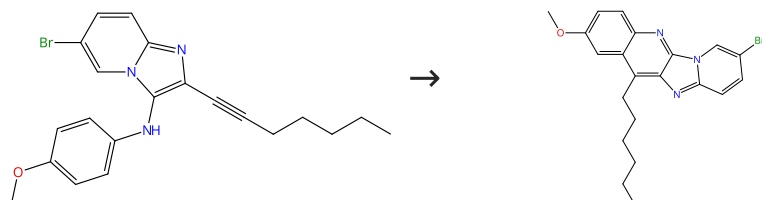
Suppliers (89)



31-614-CAS-37923778	Steps: 1 Yield: 96%	The full spectrum tuning of fluorescent molecules via a one-pot multicomponent reaction
1.1 Catalysts: Scandium triflate Solvents: Methanol; 20 min, 140 °C		By: Bedard, Nathan; et al
1.2 Reagents: Sulfuric acid; 20 min, 140 °C		Tetrahedron Letters (2023), 130, 154748.
Experimental Protocols		

Scheme 31 (1 Reaction)

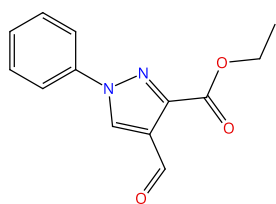
Steps: 1 Yield: 96%



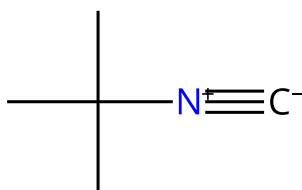
31-614-CAS-24961874	Steps: 1 Yield: 96%	Efficient Metal-Free Synthesis of Various Pyrido [2',1':2,3] imidazo- [4,5-b]quinolines
1.1 Reagents: 1,8-Diazabicyclo[5.4.0]undec-7-ene Solvents: 1,4-Dioxane; 1 h, 220 °C		By: Arnould, Mathieu; et al
Experimental Protocols		Chemistry - A European Journal (2013), 19(37), 12249-12253.

Scheme 32 (1 Reaction)

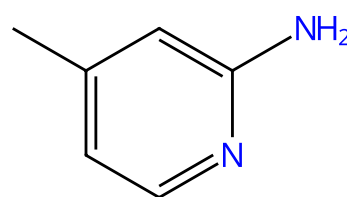
Steps: 1 Yield: 96%



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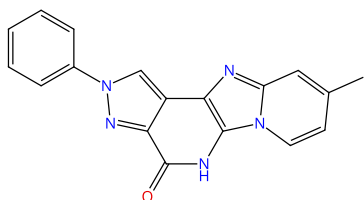


→

Suppliers (33)

Suppliers (54)

Suppliers (97)



31-614-CAS-28314996

Steps: 1 Yield: 96%

In(OTf)₃-HBF₄ Assisted Multicomponent Approach for One-Pot Synthesis of Pyrazolopyridinone Fused Imidazopyridines

By: Devi, Nisha; et al

ChemistrySelect (2016), 1(15), 4696-4703.

1.1 Catalysts: Indium triflate

Solvents: Toluene; 5 min, 120 °C

1.2 Reagents: Tetrafluoroboric acid

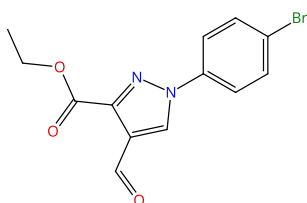
Solvents: Water; 135 min, 120 °C

1.3 Solvents: Ethyl acetate; 10 min, rt

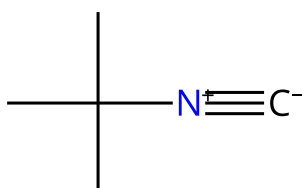
Experimental Protocols

Scheme 33 (1 Reaction)

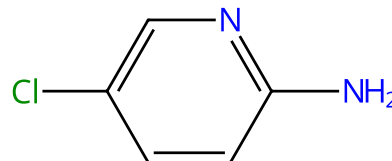
Steps: 1 Yield: 96%



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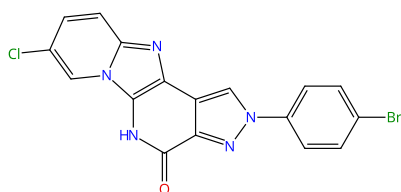


→

Suppliers (11)

Suppliers (54)

Suppliers (114)



31-614-CAS-30801469

Steps: 1 Yield: 96%

In(OTf)₃-HBF₄ Assisted Multicomponent Approach for One-Pot Synthesis of Pyrazolopyridinone Fused Imidazopyridines

By: Devi, Nisha; et al

ChemistrySelect (2016), 1(15), 4696-4703.

1.1 Catalysts: Indium triflate

Solvents: Toluene; 5 min, 120 °C

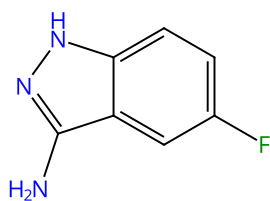
1.2 Reagents: Tetrafluoroboric acid

Solvents: Water; 135 min, 120 °C

Experimental Protocols

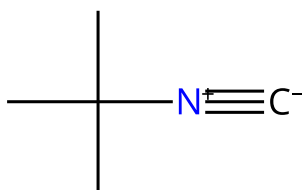
Scheme 34 (1 Reaction)

Steps: 1 Yield: 96%



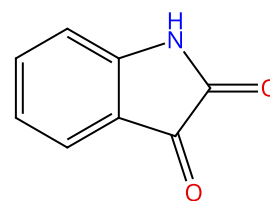
Suppliers (71)

+



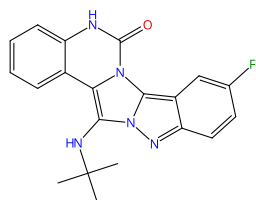
Suppliers (54)

+



Suppliers (115)

→



31-614-CAS-30731295

Steps: 1 Yield: 96%

Succinyl-β-cyclodextrin-driven synthesis of a nitrogen-fused five-ring heterocycle using GBB-based [4 + 1] cycloaddition via supramolecular host-guest interactions

1.1 Catalysts: β-Cyclodextrin, 6^A-(hydrogen butanedioate)
Solvents: Water; 30 min, 100 °C

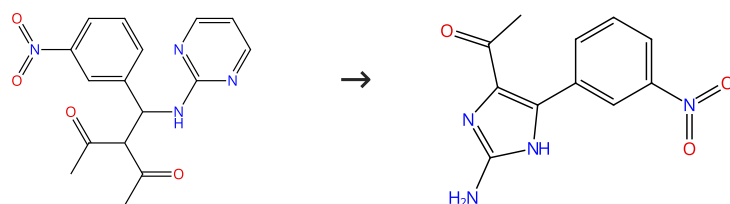
Experimental Protocols

By: Shinde, Vijay Vilas; et al

Tetrahedron (2019), 75(6), 778-783.

Scheme 35 (1 Reaction)

Steps: 1 Yield: 95%



31-614-CAS-26394602

Steps: 1 Yield: 95%

One-Pot Access towards 4,5-Disubstituted 2-Amino-1H-imidazoles Starting from Mannich Substrates and their Transformation Utilities

1.1 Reagents: 2-Iodo-1*H*-isoindole-1,3(2*H*)-dione, 2-Iodoxybenzoic acid
Solvents: Dimethylacetamide; 30 min, 80 °C; 80 °C → rt

1.2 Reagents: Sodium carbonate, Hydroxylamine hydrochloride; 15 min, rt; 16 h, 50 °C

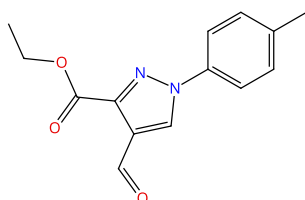
Experimental Protocols

By: Makra, Zsafia; et al

European Journal of Organic Chemistry (2020), 2020(46), 7184-7196.

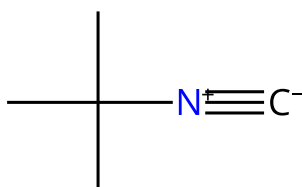
Scheme 36 (1 Reaction)

Steps: 1 Yield: 95%



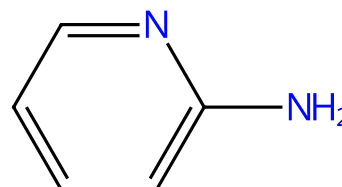
Suppliers (18)

+



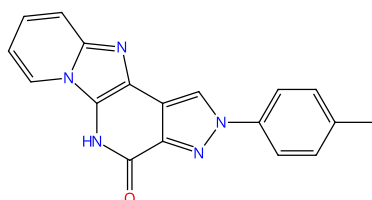
Suppliers (54)

+



Suppliers (89)

→



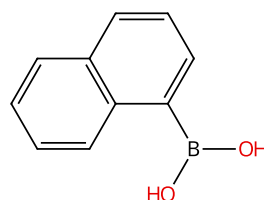
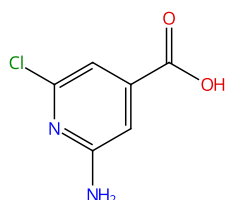
31-614-CAS-29737972	Steps: 1 Yield: 95%	In(OTf) ₃ -HBF ₄ Assisted Multicomponent Approach for One-Pot Synthesis of Pyrazolopyridinone Fused Imidazopyridines
1.1 Catalysts: Indium triflate Solvents: Toluene; 5 min, 120 °C		By: Devi, Nisha; et al
1.2 Reagents: Tetrafluoroboric acid Solvents: Water; 135 min, 120 °C		ChemistrySelect (2016), 1(15), 4696-4703.
Experimental Protocols		

Scheme 37 (1 Reaction)

Steps: 1 Yield: 95%

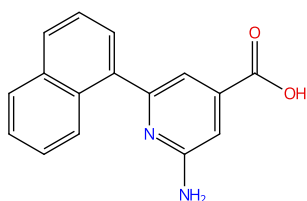
3053724-30-1

Image Not Available



Suppliers (61)

Suppliers (99)

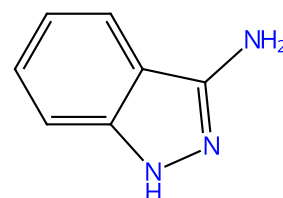
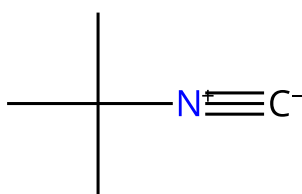
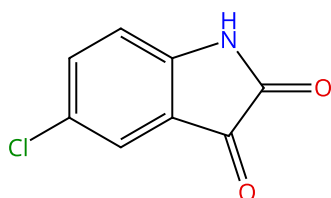


DNA-encoded library conjugates

31-614-CAS-41796238	Steps: 1 Yield: 95%	Groebke-Blackburn-Bienayme Reaction for DNA-Encoded Library Technology
1.1 Reagents: Diisopropylethylamine, <i>O</i> -(7-Azabenzotriazol-1-yl)- <i>N,N,N,N</i> -tetramethyluronium hexafluorophosphate Solvents: Dimethylacetamide, Water; overnight, pH 9.5, 40 °C		By: An, Yujin; et al
1.2 Reagents: Cesium hydroxide Catalysts: Palladate(1-), [2'-(amino-κ <i>M</i>)[1,1'-biphenyl]-2-yl-κ <i>C</i>]chloro[2'-(dicyclohexylphosphino-κ <i>P</i>)-2,6-dimethoxy[1,1'-biphenyl]-3-sulfonato]-, sodium (1:1) Solvents: 1,4-Dioxane, Dimethylacetamide, Water; 15 min, 80 °C; 10 min, rt		Organic Letters (2023), 25(24), 4445-4450.
1.3 Reagents: Sodium diethyldithiocarbamate Solvents: Water; 30 min, 60 °C		
Experimental Protocols		

Scheme 38 (1 Reaction)

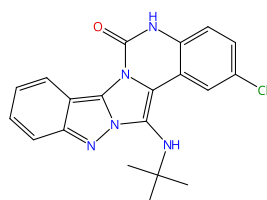
Steps: 1 Yield: 95%



Suppliers (82)

Suppliers (54)

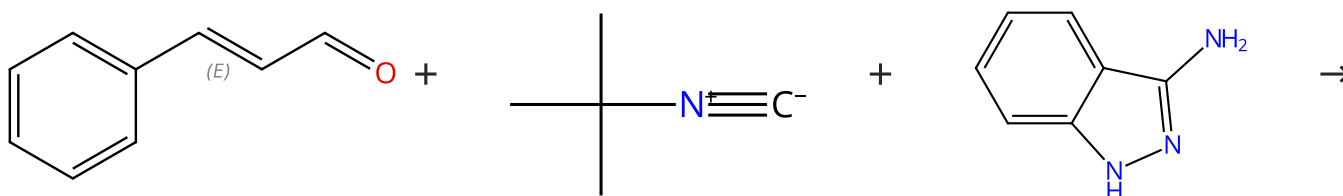
Suppliers (77)



31-614-CAS-30607649	Steps: 1 Yield: 95%	Succinyl-β-cyclodextrin-driven synthesis of a nitrogen-fused five-ring heterocycle using GBB-based [4 + 1] cycloaddition via supramolecular host-guest interactions
1.1 Catalysts: β-Cyclodextrin, 6 ^A -(hydrogen butanedioate) Solvents: Water; 40 min, 100 °C		By: Shinde, Vijay Vilas; et al
Experimental Protocols		Tetrahedron (2019), 75(6), 778-783.

Scheme 39 (1 Reaction)

Steps: 1 Yield: 95%

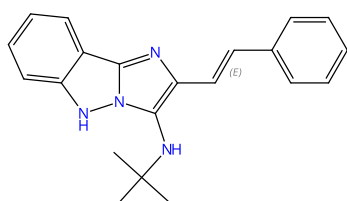


Double bond geometry shown

Suppliers (109)

Suppliers (54)

Suppliers (77)

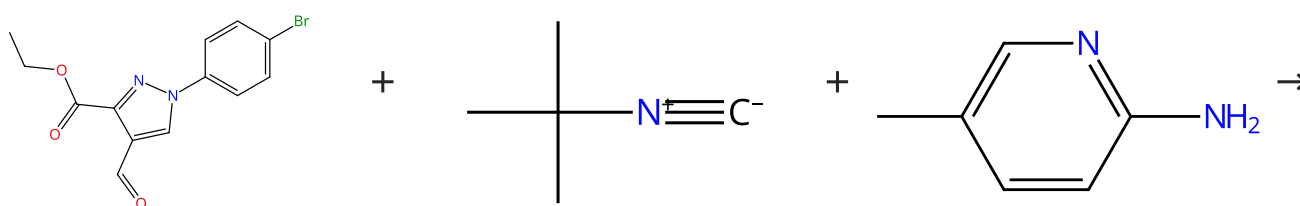


Double bond geometry shown

31-614-CAS-26794534	Steps: 1 Yield: 95%	An approach towards the synthesis of novel fused nitrogen tricyclic heterocyclic scaffolds via GBB reaction
1.1 Catalysts: Acetic acid Solvents: Ethanol; 4 h, rt		By: Balwe, Sandip Gangadhar; et al
Experimental Protocols		Organic & Biomolecular Chemistry (2018), 16(8), 1287-1296.

Scheme 40 (1 Reaction)

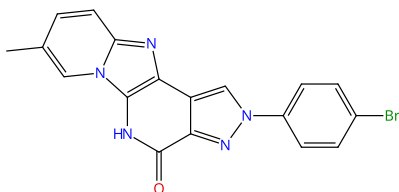
Steps: 1 Yield: 95%



Suppliers (11)

Suppliers (54)

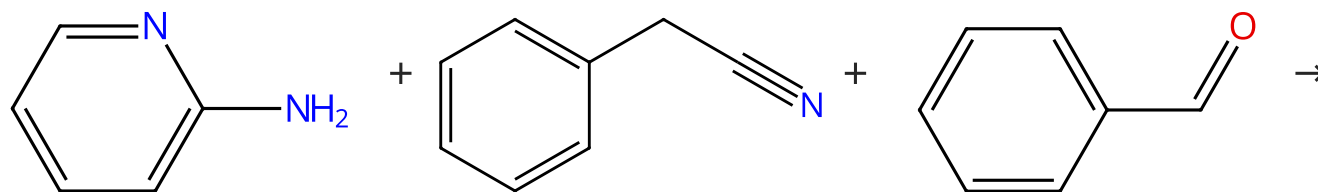
Suppliers (106)



31-614-CAS-27341344	Steps: 1 Yield: 95%	In(OTf) ₃ -HBF ₄ Assisted Multicomponent Approach for One-Pot Synthesis of Pyrazolopyridinone Fused Imidazopyridines
1.1 Catalysts: Indium triflate Solvents: Toluene; 5 min, 120 °C		By: Devi, Nisha; et al
1.2 Reagents: Tetrafluoroboric acid Solvents: Water; 135 min, 120 °C		ChemistrySelect (2016), 1(15), 4696-4703.
Experimental Protocols		

Scheme 41 (1 Reaction)

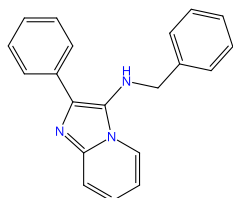
Steps: 1 Yield: 95%



Suppliers (89)

Suppliers (48)

Suppliers (69)



Suppliers (2)

31-614-CAS-30259218

Steps: 1 Yield: 95%

Visible-light-activated C-C and C-N bond formation in the synthesis of imidazo[1,2-a]pyridines and imidazo[2,1-b]thiazoles under catalyst and solvent-free conditions

By: Shivhare, Km Neha; et al

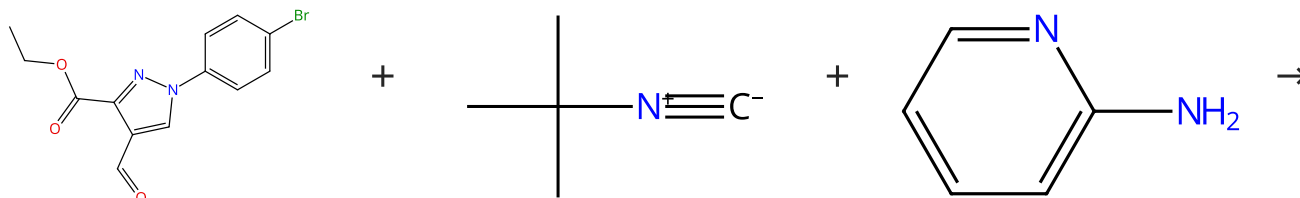
New Journal of Chemistry (2018), 42(20), 16591-16601.

1.1 3 h

Experimental Protocols

Scheme 42 (1 Reaction)

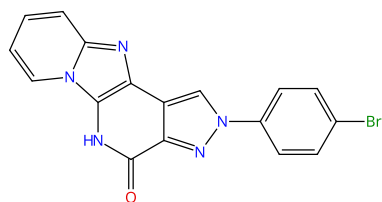
Steps: 1 Yield: 95%



Suppliers (11)

Suppliers (54)

Suppliers (89)



31-614-CAS-28873683

Steps: 1 Yield: 95%

In(OTf)₃-HBF₄ Assisted Multicomponent Approach for One-Pot Synthesis of Pyrazolopyridinone Fused Imidazopyridines

By: Devi, Nisha; et al

ChemistrySelect (2016), 1(15), 4696-4703.

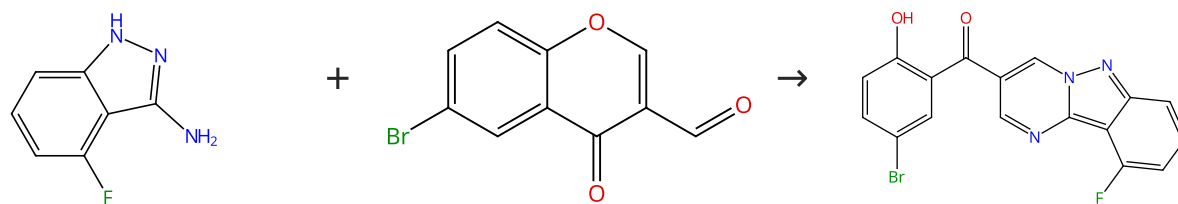
1.1 Catalysts: Indium triflate
Solvents: Toluene; 5 min, 120 °C

1.2 Reagents: Tetrafluoroboric acid
Solvents: Water; 135 min, 120 °C

Experimental Protocols

Scheme 43 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (69)

Suppliers (78)

31-614-CAS-26266606

Steps: 1 Yield: 95%

An approach towards the synthesis of novel fused nitrogen tricyclic heterocyclic scaffolds via GBB reaction

By: Balwe, Sandip Gangadhar; et al

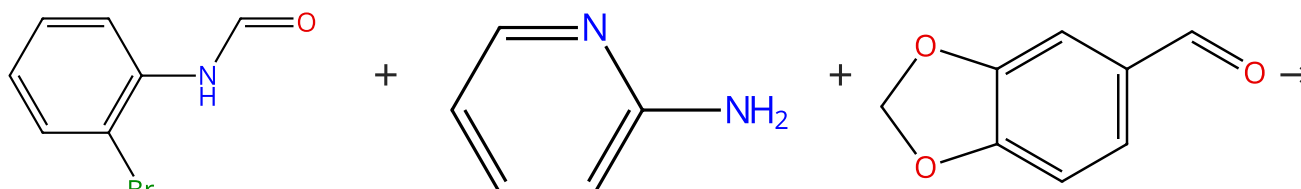
Organic & Biomolecular Chemistry (2018), 16(8), 1287-1296.

1.1 Catalysts: Acetic acid
Solvents: Ethanol; 1.5 h, rt

Experimental Protocols

Scheme 44 (1 Reaction)

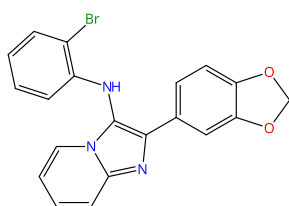
Steps: 1 Yield: 95%



Suppliers (36)

Suppliers (89)

Suppliers (42)



31-614-CAS-24623389

Steps: 1 Yield: 95%

Microwave-assisted Groebke-Blackburn-Bienayme multicomponent reaction to synthesize imidazo fused heterocycles via in-situ generated isocyanides from N-formylamines: An undergraduate organic laboratory experiment

By: Kaur, Manpreet; et al

Journal of Heterocyclic Chemistry (2022), 59(2), 319-328.

1.1 Catalysts: Triethylamine, Triphenylphosphine, Iodine
Solvents: Dichloromethane; rt; 30 min, 60 °C

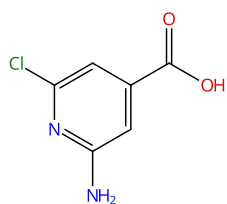
Experimental Protocols

Scheme 45 (1 Reaction)

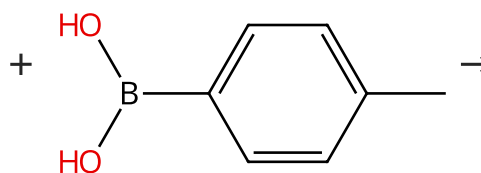
Steps: 1 Yield: 95%

3053724-30-1

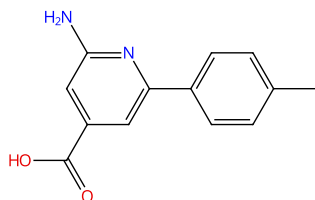
Image Not Available +



Suppliers (61)



Suppliers (114)



DNA-encoded library conjugates

31-614-CAS-41796229

Steps: 1 Yield: 95%

- 1.1 **Reagents:** Diisopropylethylamine, *O*-(7-Azabenzotriazol-1-yl)-*N,N,N,N*-tetramethyluronium hexafluorophosphate
Solvents: Dimethylacetamide, Water; overnight, pH 9.5, 40 °C
- 1.2 **Reagents:** Cesium hydroxide
Catalysts: Palladate(1-), [2'-(amino- κM)[1,1'-biphenyl]-2-yl- κC]chloro[2'-(dicyclohexylphosphino- κP)-2,6-dimethoxy[1,1'-biphenyl]-3-sulfonato]-, sodium (1:1)
Solvents: 1,4-Dioxane, Dimethylacetamide, Water; 15 min, 80 °C; 10 min, rt
- 1.3 **Reagents:** Sodium diethyldithiocarbamate
Solvents: Water; 30 min, 60 °C

Experimental Protocols

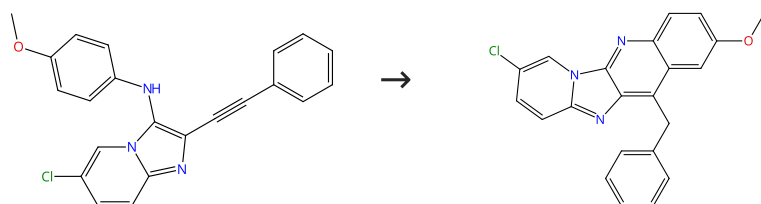
Groebke-Blackburn-Bienayme Reaction for DNA-Encoded Library Technology

By: An, Yujin; et al

Organic Letters (2023), 25(24), 4445-4450.

Scheme 46 (1 Reaction)

Steps: 1 Yield: 95%



31-614-CAS-25074128

Steps: 1 Yield: 95%

- 1.1 **Reagents:** 1,8-Diazabicyclo[5.4.0]undec-7-ene
Solvents: 1,4-Dioxane; 10 min, 220 °C

Experimental Protocols

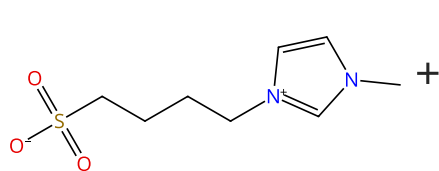
Efficient Metal-Free Synthesis of Various Pyrido [2',1':2,3]imidazo- [4,5-b]quinolines

By: Arnould, Mathieu; et al

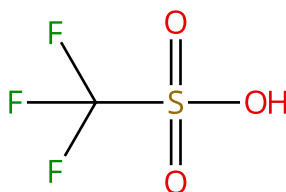
Chemistry - A European Journal (2013), 19(37), 12249-12253.

Scheme 47 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (41)



Suppliers (77)



Multi-component
structure image
available in CAS
SciFinder

Suppliers (42)

31-614-CAS-34387881

Steps: 1 Yield: 95%

**Groebe-Blackburn-Bienayme Multicomponent Reaction
Catalysed by Reusable Bronsted acidic ionic Liquids**

1.1 Solvents: Dichloromethane; 0 °C; 24 h, rt

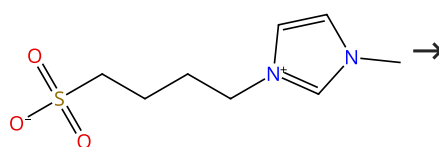
Experimental Protocols

By: Anjos, Nicolas S.; et al

European Journal of Organic Chemistry (2022), 2022(40),
e202200615.

Scheme 48 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (41)

Multi-component
structure image
available in CAS
SciFinder

Suppliers (2)

31-614-CAS-34387876

Steps: 1 Yield: 95%

**Groebe-Blackburn-Bienayme Multicomponent Reaction
Catalysed by Reusable Bronsted acidic ionic Liquids**

1.1 Reagents: Tetrafluoroboric acid
Solvents: Water; 12 h, rt

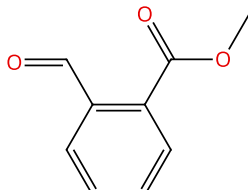
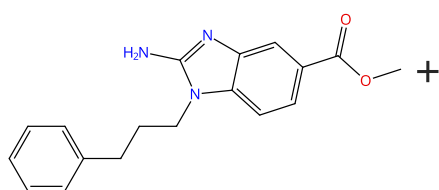
Experimental Protocols

By: Anjos, Nicolas S.; et al

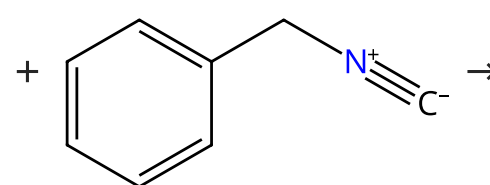
European Journal of Organic Chemistry (2022), 2022(40),
e202200615.

Scheme 49 (1 Reaction)

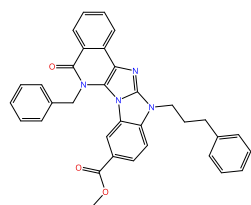
Steps: 1 Yield: 95%



Suppliers (74)

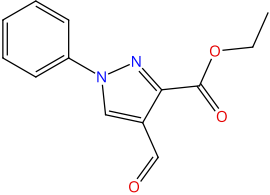


Suppliers (65)



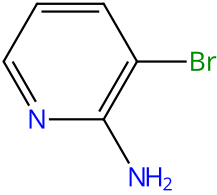
31-614-CAS-27218463 Steps: 1 Yield: 95% 1.1 Reagents: Sulfuric acid magnesium salt (1: 1) Catalysts: Piperidine Solvents: Dichloromethane; 12 h, reflux 1.2 Catalysts: Piperidine Solvents: Toluene; 12 h, heated	A Telescoping Synthesis of Chimeric Polyhete rocycles through a Piperidine-Mediated Multicomponent Reaction By: Lee, Chia-Hsin; et al European Journal of Organic Chemistry (2013), 2013(11), 2201-2208.
Experimental Protocols	

Scheme 50 (1 Reaction) Steps: 1 Yield: 95%



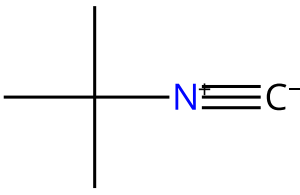
 Suppliers (33)

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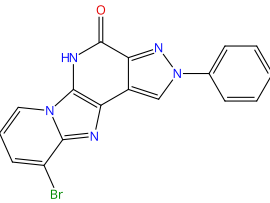
 Suppliers (95)

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 Suppliers (54)

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31-614-CAS-27400428 Steps: 1 Yield: 95% 1.1 Catalysts: Indium triflate Solvents: Toluene; 5 min, 120 °C 1.2 Reagents: Tetrafluoroboric acid Solvents: Water; 135 min, 120 °C 1.3 Solvents: Ethyl acetate; 10 min, rt	In(OTf)₃-HBF₄ Assisted Multicomponent Approach for One-Pot Synthesis of Pyrazolopyridinone Fused Imidazopyridines By: Devi, Nisha; et al ChemistrySelect (2016), 1(15), 4696-4703.
Experimental Protocols	